

2.1

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$ $\sum y_i = 10$ $\sum (x_i - \bar{x}) = 0$ $\sum (x_i - \bar{x})^2 = 10$ $\sum (y_i - \bar{y}) = 0$					$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

$$a. \bar{x} = \frac{\sum x_i}{N} = 1, \bar{y} = \frac{\sum y_i}{N} = 2$$

$$b. b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8, b_1 = \bar{y} - b_2 \bar{x} = 1.2 \quad \text{where } \bar{x} = \frac{\sum x_i}{N}, \bar{y} = \frac{\sum y_i}{N} \text{ are the sample means of observations on } y \text{ and } x$$

$$c. \sum x_i^2 = 15, \sum x_i y_i = 18 \quad \sum (x_i - \bar{x})^2 = \sum x_i^2 - N \bar{x}^2 = 15 - 5 \cdot 1^2 = 10$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N \bar{x} \bar{y} = 18 - 5 \cdot 1 \cdot 2 = 8$$

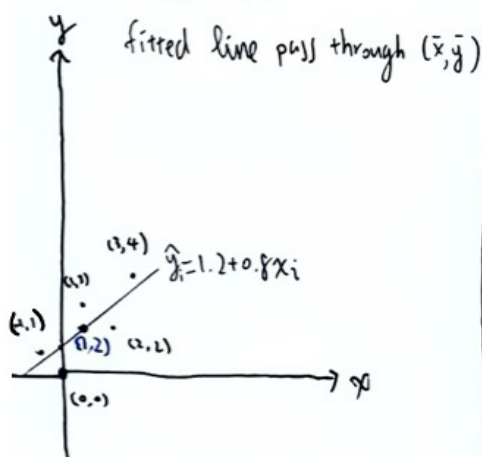
x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i = 5$ $\sum y_i = 10$ $\sum \hat{y}_i = 10$ $\sum \hat{e}_i = 0$ $\sum \hat{e}_i^2 = 3.6$ $\sum x_i \hat{e}_i = 0$					

$$s_y^2 = \frac{\sum (y_i - \bar{y})^2}{N-1} = 2.5 \quad s_x^2 = \frac{\sum (x_i - \bar{x})^2}{N-1} = 2.5$$

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N-1} = 2 \quad r_{xy} = \frac{s_{xy}}{s_x s_y} = 0.8$$

$$(V_x = 100 \frac{s_x}{\bar{x}} = 158.11\% \text{ Median of } x = 1)$$

$$e, f. (\bar{x}, \bar{y}) = (1, 2)$$



$$g. \bar{y} = b_1 + b_2 \bar{x}$$

$$2 = 1.2 + 0.8 \cdot 1$$

$$i. \hat{\sigma}^2 = \frac{\sum (y_i - b_1 - b_2 x_i)^2}{N-2} = \frac{3.6}{3} = 1.2$$

$$h. \hat{y} = \bar{y} \Rightarrow \frac{\sum \hat{y}_i}{N} = \bar{y}$$

$$\frac{10}{5} = 2$$

$$j. \hat{\text{var}}(b_2 | x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = 0.12$$

$$se(b_2) = \sqrt{0.12} = 0.3464$$

2.22

(a) Class with regular-size has a mean TOTALSCORE=916.442 while the small-size one has a higher mean TOTALSCORE than regular-size one by 12.175 points.

p-value=0.0236=> It is significant that small class has higher mean TOTALSCORE than regular one. We can conclude that small class size has positive effect on learning.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	916.442	3.675	249.396	<2e-16 ***
small	12.175	5.369	2.268	0.0236 *

(b) For READSCORE, small class has a higher mean READSCORE than regular one by 6.924 points. Since p-value=0.00151, It is significant that small class has a higher mean READSCORE. We can conclude that small class size has positive effect on reading.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	432.665	1.488	290.760	< 2e-16 ***
small	6.924	2.174	3.185	0.00151 **

For MATHSCORE, small class has a higher mean MATHSCORE than regular one by 5.251 points. Since p-value=0.143, It is not significant that small class has a higher mean MATHSCORE. We can't conclude that small class size has positive effect on learning math.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	483.777	2.449	197.545	<2e-16 ***
small	5.251	3.578	1.467	0.143

(c) Class with aide has a higher mean TOTALSCORE than the one without aide by 4.306 points. Since p-value=0.389, it is not statistically significant that aide has positive effect on learning.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	916.442	3.559	257.529	<2e-16 ***
aide	4.306	4.994	0.862	0.389

(d) For READSCORE, class with aide has a higher mean READSCORE than regular one by 2.871 points. Since p-value=0.161, It is not significant that class with aide has a higher mean READSCORE. We cannot conclude that aide has positive effect on reading.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	432.665	1.460	296.341	<2e-16 ***
aide	2.871	2.049	1.401	0.161

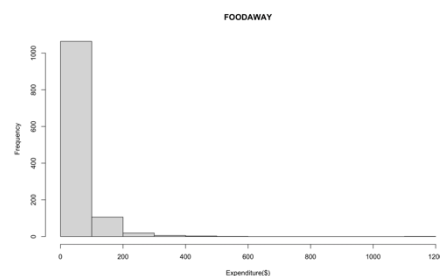
For MATHSCORE, class with aide has a higher mean MATHSCORE than regular one by 1.435 points. Since p-value=0.664, It is not significant that class with aide has a higher mean MATHSCORE. We cannot conclude that aide has positive effect on learning math.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	483.777	2.355	205.464	<2e-16 ***
aide	1.435	3.304	0.434	0.664

2.25

(a)



```
> summary(cex5_small$foodaway)
Min. 1st Qu. Median Mean 3rd Qu. Max.
 0.00  12.04  32.55  49.27  67.50 1179.00
> quantile(cex5_small$foodaway, probs=c(0.25,0.75))
 25%    75%
12.0400 67.5025
```

(b) member with an advanced degree:

```
summary(adv_group$foodaway)
Min. 1st Qu. Median Mean 3rd Qu. Max.
 0.00  21.67  48.15  73.15  90.00 1179.00
```

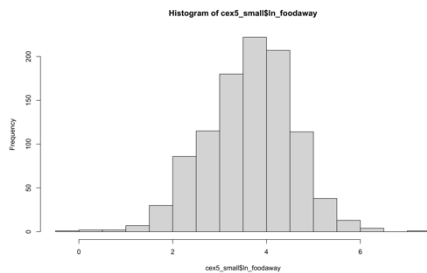
member with a college degree:

```
summary(col_group$foodaway)
Min. 1st Qu. Median Mean 3rd Qu. Max.
 0.00  14.44  36.11  48.60  68.67 416.11
```

With no advanced or college degree:

```
summary(none_group$foodaway)
Min. 1st Qu. Median Mean 3rd Qu. Max.
 0.00   9.63  26.02  39.01  52.65 437.78
```

(c) Since some home have FOODAWAY=0, $\ln(\text{FOODAWAY})$ will ignore those observations. This is why FOODAWAY and $\ln(\text{FOODAWAY})$ have different numbers of observations.



Observations in FOODAWAY (a): 1200

Observations in $\ln(\text{FOODAWAY})$ (c): 1022

```
summary(cex5_small$ln_foodaway)
```

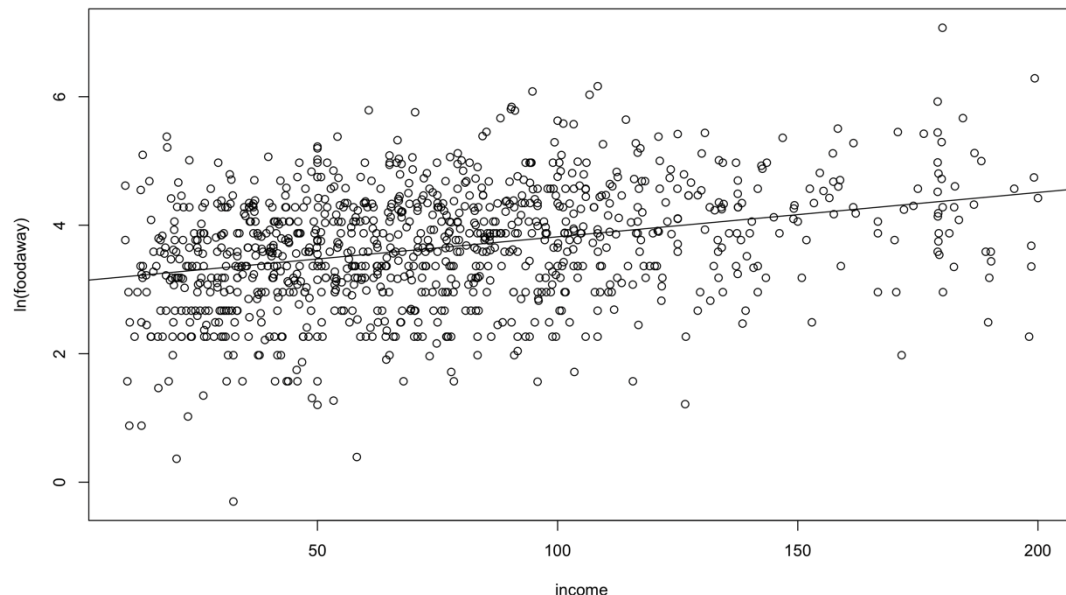
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-Inf	2.488	3.483	-Inf	4.212	7.072

(d) Due to log-linear model, $\beta_2=0.0069$ means that when income increases 1 unit, the FOODAWAY will increase 0.69%. Since $p\text{-value} < 2e-16$, it is significant that income has positive effect on FOODAWAY expenditure.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.1293004	0.0565503	55.34	<2e-16 ***
income	0.0069017	0.0006546	10.54	<2e-16 ***

(e)



(f) The residuals do not seem completely random. Instead, they exhibit a clear pattern of heteroskedasticity.

As income increases, the spread of the residuals becomes wider. This suggests that the error variance is not constant, violating one of the key Gauss-Markov assumptions (Homoskedasticity).

